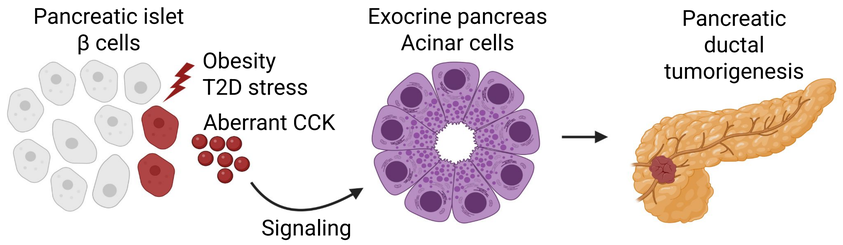
# Learning cellular and molecular mechanisms of pancreatic cancer

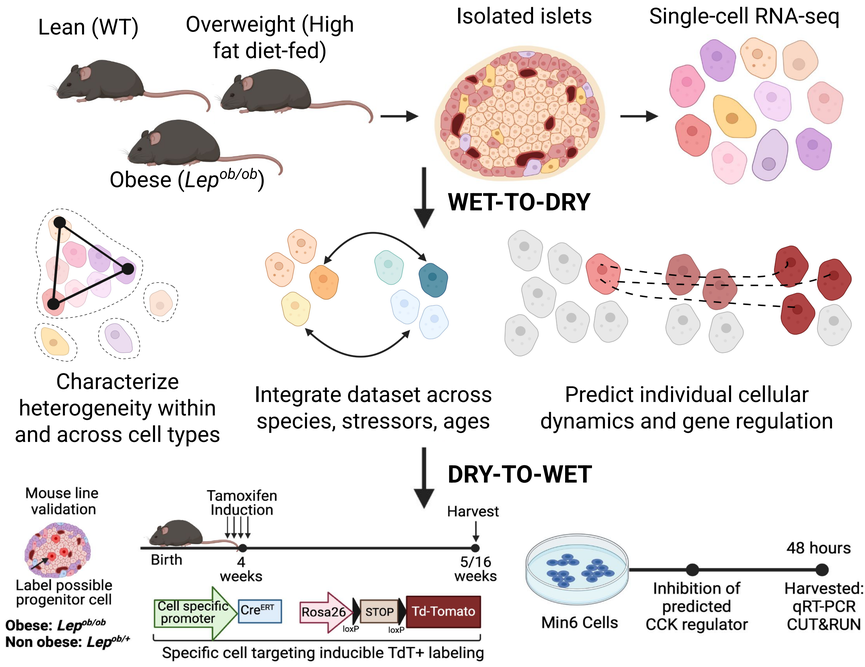

**AARTHI VENKAT[1,#,+], CATHY GARCIA[2,#], … , SMITA KRISHNASWAMY[1,2,3,§], MANDAR DEEPAK MUZUMDAR[2,§]**

[1]Eric and Wendy Schmidt Center, Broad Institute of MIT and Harvard
[2]Department of Genetics, Yale School of Medicine
[3]Department of Computer Science, Yale University
[#§]Contributed equally [+]**Contact: avenkat@broadinstitute.org**


## ENDOCRINE-EXOCRINE SIGNALING DRIVES OBESITY-ASSOCIATED PANCREATIC CANCER[1]




## WHAT ARE THE CELLULAR AND MOLECULAR MECHANISMS BY WHICH ISLETS ADAPT IN RESPONSE TO OBESITY TO EXPRESS CCK?

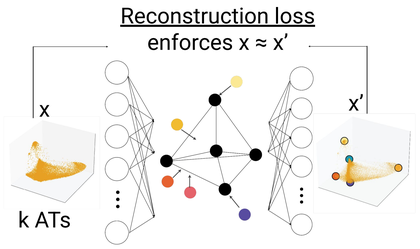




1. Chung, K.M,. et al. Endocrine-Exocrine Signaling Drives Obesity-Associated Pancreatic Ductal Adenocarcinoma. *Cell*.
2. Venkat, A., Youlten, S.E., San Juan, B.P., et al. AAnet resolves a continuum of spatially- localized cell states to unveil intratumoral heterogeneity. *Cancer Discovery*.
3. Amodio, M., et al. Single-cell multi-modal GAN reveals spatial patterns in single-cell data from triple- negative breast cancer. *Cell Patterns*.
4. Tong, A. et al. TrajectoryNet: A Dynamic Optimal Transport Network for Modeling Cellular Dynamics. *ICML*.

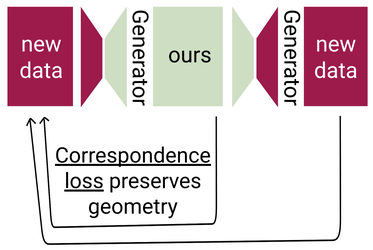

**Read more about this work "Beta cells are essential drivers of PDAC development" here**

## AANET[2] LEARNS SIMPLEX TO MAP CELLS BY ARCHETYPES



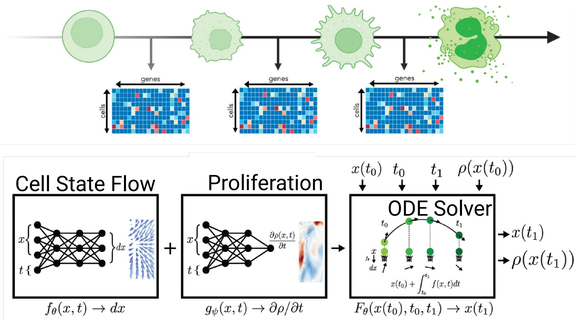

Archetypal loss enforces points to be within k-1 simplex

## SCMMGAN[3] MAPS VIA ADVERSARIAL LEARNING

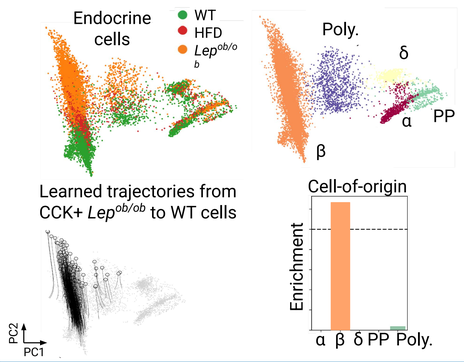



Reconstruction loss enforces cycle consistency

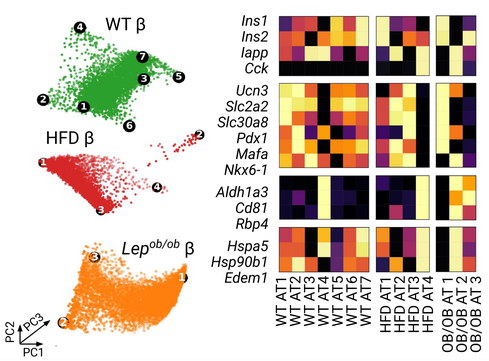
## TRAJECTORYNET[4] LEARNS DYNAMICS FROM SNAPSHOTS VIA DYNAMIC OT



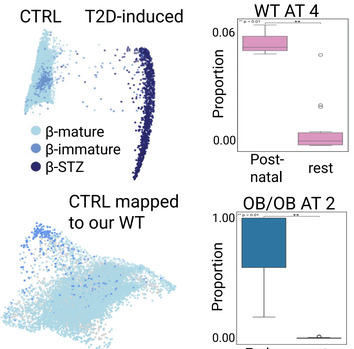

## CCK+ CELLS ARISE FROM PREEXISTING β CELLS



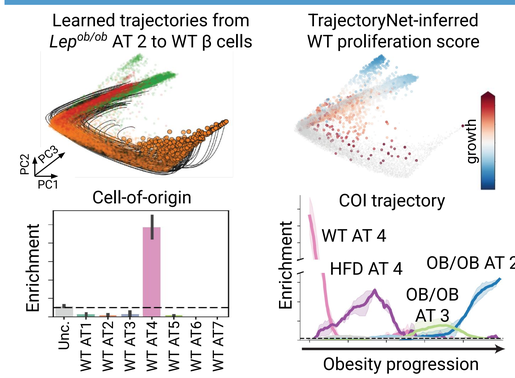

## β CELL ADAPTATION SHOWS SIGNATURES OF IMMATURITY

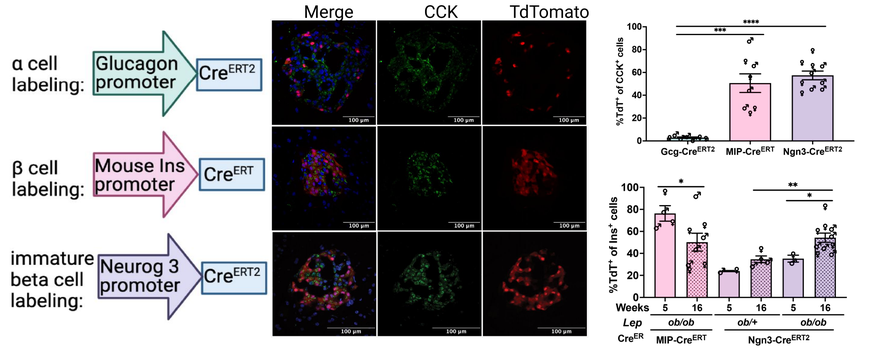



## IMMATURE ARCHETYPES EXIST IN T2D, P16 MICE




## IMMATURE β CELLS EXPAND AND ADAPT TO EXPRESS CCK




## C-JUN REGULATES CCK EXPRESSION VIA PUTATIVE 3' ENHANCER

